



PROJECT REPORT: GARBAGE CLASSIFICATION

COURSE: DEEP LEARNING

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Abstract

Waste management is a major environmental challenge in modern societies. Manual waste sorting is time-consuming, expensive, and error-prone. This project presents an automated garbage classification system using image-based Machine Learning and Deep Learning techniques. A six-class garbage dataset was used to classify waste into categories such as cardboard, glass, metal, paper, plastic, and trash. A Machine Learning baseline (Logistic Regression), a Convolutional Neural Network (CNN), and an advanced Transfer Learning model (MobileNetV2) were implemented and evaluated. Experimental results show that deep learning, especially transfer learning, significantly outperforms traditional machine learning methods, demonstrating the effectiveness of automated waste classification systems.

1. Introduction

With the rapid increase in urbanization and population growth, waste generation has increased significantly. Proper waste segregation is essential for recycling and environmental sustainability. Traditional manual waste sorting methods are inefficient and require significant human effort. Recent advancements in Artificial Intelligence and Deep Learning have enabled automated image-based classification systems that can assist in waste management. This project focuses on developing an image classification model capable of automatically identifying different types of garbage using machine learning and deep learning techniques.

2. Literature Review

Previous research has shown that Convolutional Neural Networks (CNNs) perform exceptionally well in image classification tasks. Studies on waste classification have utilized CNN architectures such as VGG, ResNet, and MobileNet to achieve high accuracy. Transfer learning has been widely adopted to improve performance when datasets are limited. MobileNetV2, in particular, is known for its lightweight architecture and efficiency, making it suitable for real-world deployment. These studies inspired the design and implementation of the models used in this project.

3. Dataset Description

The dataset used in this project is a garbage classification dataset obtained from Kaggle. It contains images belonging to six categories: Cardboard, Glass, Metal, Paper, Plastic, Trash. All images were organized into class-wise folders. The dataset was split into training and testing sets using TensorFlow's image data generator.

4. Data Preprocessing

Several preprocessing steps were applied before model training: Image resizing to 128×128 pixels, Pixel normalization (scaling values between 0 and 1), Label encoding using categorical (one-hot) encoding, Train and test dataset splitting. These steps ensured consistency in input data and improved model convergence.

5. Methodology

5.1 Machine Learning Baseline

A Logistic Regression model was implemented as a baseline for benchmarking. Images were flattened into one-dimensional vectors before training. This model provided a reference point to compare the performance of deep learning models.

5.2 Deep Learning Model (CNN)

A Convolutional Neural Network (CNN) was designed using multiple convolutional and pooling layers followed by dense layers. The CNN learned spatial features directly from the images and achieved significantly better performance than the machine learning baseline.

5.3 Advanced Enhancement: Transfer Learning

To further improve performance, transfer learning was applied using MobileNetV2 pre-trained on ImageNet. The base model was frozen, and custom classification layers were added. This approach reduced training time and improved accuracy by leveraging learned features from a large-scale dataset.

6. Model Architecture

- **CNN Model:**
 - Convolutional layers with ReLU activation
 - Max pooling layers
 - Fully connected dense layers
 - Softmax output layer
- **MobileNetV2 Model:**
 - Pre-trained MobileNetV2 backbone
 - Global average pooling
 - Dense layers for classification

7. Evaluation Metrics

The following evaluation metrics were used: Accuracy, Precision, Recall, F1Score, Confusion Matrix. These metrics provided a comprehensive evaluation of model performance across all classes.

8. Experimental Results

The Logistic Regression baseline achieved relatively low accuracy due to its inability to capture spatial features. The CNN model significantly improved classification performance. The MobileNetV2 transfer learning model achieved the highest accuracy and F1-score, demonstrating the effectiveness of transfer learning for image classification tasks.

9. Discussion

Results indicate that deep learning models outperform traditional machine learning approaches for image-based garbage classification. Transfer learning further enhances performance, especially when working with limited datasets. The confusion matrix analysis showed that most misclassifications occurred between visually similar classes such as plastic and metal.

10. Conclusion

This project successfully demonstrates an automated garbage classification system using machine learning and deep learning techniques. The results highlight the superiority of CNNs and transfer learning models over traditional methods. Such systems can play a vital role in improving waste management efficiency and supporting environmental sustainability. Future work may include real-time deployment and expansion to additional waste categories.